

5.1 INTRODUCTION

Magnetic phenomena are universal in nature. Vast, distant galaxies, the tiny invisible atoms, humans and beasts all are permeated through and through with a host of magnetic fields from a variety of sources. The earth's magnetism predates human evolution. The word magnet is derived from the name of an island in Greece called magnesia where magnetic ore deposits were found, as early as 600 BC. In the previous chapter we have learned that moving charges or electric currents produce magnetic fields. This discovery, which was made in the early part of the nineteenth century is credited to Oersted, Ampere, Biot and Savart, among others. In the present chapter, we take a look at magnetism as a subject in its own right. Some of the commonly known ideas regarding magnetism are: (i) The earth behaves as a magnet with the magnetic field pointing approximately from the geographic south to the north. (ii) When a bar magnet is freely suspended, it points in the north-south direction. The tip which points to the geographic north is called the north pole and the tip which points to the geographic south is called the south pole of the magnet. (iii) There is a repulsive force when north poles (or south poles) of two magnets are brought close together. Conversely, there is an attractive force between the north pole of one magnet and the south pole of the other. (iv) We cannot isolate the north, or south pole of a magnet. If a bar magnet is broken into two halves, we get two similar bar magnets with somewhat weaker properties. Unlike electric charges, isolated magnetic north and south poles known as magnetic monopoles do not exist. (v) It is possible to make magnets out of iron and its alloys. We begin with a description of a bar magnet and its behaviour in an external magnetic field. We describe Gauss's law of magnetism. We then follow it up with an account of the earth's magnetic field. We next describe how materials can be classified on the basis of their magnetic properties. We describe para-, dia-, and ferromagnetism. We conclude with a section on electromagnets and permanent magnets.

5.2 THE BAR MAGNET

One of the earliest childhood memories of the famous physicist Albert Einstein was that of a magnet gifted to him by a relative. Einstein was fascinated, and played endlessly with it. He wondered how the magnet could affect objects such as nails or pins placed away from it and not in any way connected to it by a spring or string. We begin our study by examining iron filings sprinkled on a sheet of glass placed over a short bar magnet. The arrangement of iron filings is shown in Fig. 5.1. The pattern of iron filings suggests that the magnet has two poles similar to the positive and negative charge of an electric dipole. As mentioned in the introductory section, one pole is designated the North pole and the other, the South pole. When suspended freely, these poles point approximately towards the geographic north and south poles, respectively. A similar pattern of iron filings is observed around a current carrying solenoid.

5.2.1 The magnetic field lines

The pattern of iron filings permits us to plot the magnetic field lines*. This is shown both for the bar-magnet and the current-carrying solenoid in Fig. 5.2. For comparison refer to the Chapter 1, Figure 1.17(d). Electric field lines of an electric dipole are also displayed in Fig. 5.2(c). The magnetic field lines are a visual and intuitive realisation of the magnetic field. Their properties are: (i) The magnetic field lines of a magnet (or a solenoid) form continuous closed loops. This is unlike the electric dipole where

these field lines begin from a positive charge and end on the negative charge or escape to infinity. (ii) The tangent to the field line at a given point represents the direction of the net magnetic field B at that point. (iii) The larger the number of field lines crossing per unit area, the stronger is the magnitude of the magnetic field B . In Fig. 5.2(a), B is larger around region ii than in region i. (iv) The magnetic field lines do not intersect, for if they did, the direction of the magnetic field would not be unique at the point of intersection. One can plot the magnetic field lines in a variety of ways. One way is to place a small magnetic compass needle at various positions and note its orientation. This gives us an idea of the magnetic field direction at various points in space.

5.2.2 Bar magnet as an equivalent solenoid

In the previous chapter, we have explained how a current loop acts as a magnetic dipole (Section 4.10). We mentioned Ampere's hypothesis that all magnetic phenomena can be explained in terms of circulating currents.

The resemblance of magnetic field lines for a bar magnet and a solenoid suggest that a bar magnet may be thought of as a large number of circulating currents in analogy with a solenoid. Cutting a bar magnet in half is like cutting a solenoid. We get two smaller solenoids with weaker magnetic properties. The field lines remain continuous, emerging from one face of the solenoid and entering into the other face. One can test this analogy by moving a small compass needle in the neighbourhood of a bar magnet and a current-carrying finite solenoid and noting that the deflections of the needle are similar in both cases. To make this analogy more firm we calculate the axial field of a finite solenoid depicted in Fig. 5.3 (a). We shall demonstrate that at large distances this axial field resembles that of a bar magnet.

$$B = (\mu_0 / 4\pi) * 2m/r^3 \quad (5.1).$$

This is also the far axial magnetic field of a bar magnet which one may obtain experimentally. Thus, a bar magnet and a solenoid produce similar magnetic fields. The magnetic moment of a bar magnet is thus equal to the magnetic moment of an equivalent solenoid that produces the same magnetic field.

5.2.3 The dipole in a uniform magnetic field

Let's place a small compass needle of known magnetic moment m and allowing it to oscillate in the magnetic field. This arrangement is shown in Fig. 5.3(b). The torque on the needle is [see Eq. (4.23)],

$$\tau = m \times B \quad (5.2)$$

In magnitude $\tau = mB \sin \theta$

Here, τ is restoring torque and θ is the angle between m and B . An expression for magnetic potential energy can also be obtained on lines similar to electrostatic potential energy. The magnetic potential energy U_m is given by

$$U_m = \int \tau(\theta) d\theta$$

$$\int mB \sin \theta d\theta = -mB \cos \theta$$

$$= -m \cdot B \quad (5.3)$$

We have emphasised in Chapter 2 that the zero of potential energy can be fixed at one's convenience. Taking the constant of integration to be zero means fixing the zero of potential energy at $\theta = 90^\circ$, i.e., when the needle is perpendicular to the field. Equation (5.6) shows that potential energy is minimum ($= -mB$) at $\theta = 0^\circ$ (most stable position) and maximum ($= +mB$) at $\theta = 180^\circ$ (most unstable position).

5.2.4 The electrostatic analog

Comparison of Eqs. (5.2), (5.3) and (5.6) with the corresponding equations for electric dipole (Chapter 1), suggests that magnetic field at large distances due to a bar magnet of magnetic moment m can be obtained from the equation for electric field due to an electric dipole of dipole moment p , by making the following replacements:

$$1/4\pi\epsilon_0 \rightarrow \mu_0/4\pi$$

In particular, we can write down the equatorial field (B_E) of a bar magnet at a distance r , for $r \gg l$, where l is the size of the magnet:

$$B_E = -(\mu_0 m/4\pi r^3) \quad (5.4)$$

Likewise, the axial field (B_A) of a bar magnet for $r \gg l$ is:

$$B_A = (\mu_0/4\pi) * (2m/r^3) \quad (5.5)$$

Equation (5.8) is just Eq. (5.2) in the vector form. Table 5.1 summarises the analogy between electric and magnetic dipoles.

Table 5.1 The Dipole Analogy

	Electrostatics	Magnetism
	$1/\epsilon_0$	μ_0
Dipole moment	p	m
Equatorial Field for a short dipole	$-(p/4\pi\epsilon_0 r^3)$	$-(\mu_0 m/4\pi r^3)$
Axial Field for a short dipole	$2p/4\pi\epsilon_0 r^3$	$2\mu_0 m/4\pi r^3$
External Field : torque	$p \times E$	$m \times B$
External Field : Energy	$-p \cdot E$	$-m \cdot B$

5.3 MAGNETISM AND GAUSS'S LAW

In Chapter 1, we studied Gauss's law for electrostatics. In Fig 5.3(c), we see that for a closed surface represented by i, the number of lines leaving the surface is equal to the number of lines entering it. This is consistent with the fact that no net charge is enclosed by the surface. However, in the same figure, for the closed surface ii, there is a net outward flux, since it does include a net (positive) charge. The situation is radically different for magnetic fields which are continuous and form closed loops. Examine the Gaussian surfaces represented by i or ii in Fig 5.3(a) or Fig. 5.3(b). Both cases visually demonstrate that the number of magnetic field lines leaving the surface is balanced by the number of lines entering it. The net magnetic flux is zero for both the surfaces. This is true for any closed surface.

Consider a small vector area element ΔS of a closed surface S as in Fig. 5.5. The magnetic flux through ΔS is defined as $\Delta \Phi_B = \mathbf{B} \cdot \Delta \mathbf{S}$, where $\mathbf{B}$ is the field at ΔS . We divide S into many small area elements and calculate the individual flux through each. Then, the net flux Φ_B is,

$$\Phi_B = \sum \Delta \Phi_B = \sum \mathbf{B} \cdot \Delta \mathbf{S}. \quad (5.6)$$

where 'all' stands for 'all area elements $\Delta \mathbf{S}$ '. Compare this with the Gauss's law of electrostatics. The flux through a closed surface in that case is given by

$$\sum \mathbf{E} \cdot \Delta \mathbf{S} = q/\epsilon_0$$

where q is the electric charge enclosed by the surface. The difference between the Gauss's law of magnetism and that for electrostatics is a reflection of the fact that isolated magnetic poles (also called monopoles) are not known to exist. There are no sources or sinks of $\mathbf{B}$; the simplest magnetic element is a dipole or a current loop. All magnetic phenomena can be explained in terms of an arrangement of dipoles and/or current loops. Thus, Gauss's law for magnetism is: The net magnetic flux through any closed surface is zero.

5.4 MAGNETISATION AND MAGNETIC INTENSITY

The earth abounds with a bewildering variety of elements and compounds. In addition, we have been synthesising new alloys, compounds and even elements. One would like to classify the magnetic properties of these substances. In the present section, we define and explain certain terms which will help us to carry out this exercise. We have seen that a circulating electron in an atom has a magnetic moment. In a bulk material, these moments add up vectorially and they can give a net magnetic moment which is non-zero. We define magnetisation M of a sample to be equal to its net magnetic moment per unit volume:

$$M = m_{\text{net}} / V \quad (5.7)$$

M is a vector with dimensions $L^{-1} A$ and is measured in a units of $A m^{-1}$.

Consider a long solenoid of n turns per unit length and carrying a current I . The magnetic field in the interior of the solenoid was shown to be given by

$$B_0 = \mu_0 nI \quad (5.8)$$

If the interior of the solenoid is filled with a material with non-zero magnetisation, the field inside the solenoid will be greater than B_0 . The net B field in the interior of the solenoid may be expressed as

$$B = B_0 + B_m \quad (5.9)$$

where B_m is the field contributed by the material core. It turns out that this additional field is proportional to the magnetisation M of the material and is expressed as

$$B_m = \mu_0 M \quad (5.10)$$

where μ_0

is the same constant (permittivity of vacuum) that appears in Biot-Savart's law. It is convenient to introduce another vector field H , called the magnetic intensity, which is defined by

$$H = B / \mu_0 M \quad (5.11)$$

where H has the same dimensions as M and is measured in units of $A \, m^{-1}$. Thus, the total magnetic field B is written as

$$B = \mu_0 (H + M) \quad (5.12)$$

We repeat our defining procedure. We have partitioned the contribution to the total magnetic field inside the sample into two parts: one, due to external factors such as the current in the solenoid. This is represented by H . The other is due to the specific nature of the magnetic material, namely M . The latter quantity can be influenced by external factors. This influence is mathematically expressed as

$$M = H \chi \quad (5.13)$$

where χ , a dimensionless quantity, is appropriately called the magnetic susceptibility. It is a measure of how a magnetic material responds to an external field. χ is small and positive for materials, which are called paramagnetic. It is small and negative for materials, which are termed diamagnetic. In the latter case M and H are opposite in direction. From Eqs. (5.12) and (5.13) we obtain,

$$B = \mu_0 (1 + \chi) H \quad (5.14)$$

$$= \mu_0 \mu_r H$$

$$= \mu H \quad (5.15)$$

Where, $\mu_r = 1 + \chi$, is a dimensionless quantity called the relative magnetic permeability of the substance. It is the analog of the dielectric constant in electrostatics. The magnetic permeability of the substance is μ and it has the same dimensions and units as μ_0 ;

$$\mu = \mu_0 \mu_r = \mu_0 (1 + \chi).$$

The three quantities χ , μ_0 and μ_r are interrelated and only one of them is independent. Given one, the other two may be easily determined.

5.5 MAGNETIC PROPERTIES OF MATERIALS

The discussion in the previous section helps us to classify materials as diamagnetic, paramagnetic or ferromagnetic. In terms of the susceptibility χ , a material is diamagnetic if χ is negative, para- if χ is positive and small, and ferro- if χ is large and positive.

A glance at Table 5.3 gives one a better feeling for these materials. Here ϵ is a small positive number introduced to quantify paramagnetic materials. Next, we describe these materials in some detail.

Table 5.3

Diamagnetic	Paramagnetic	Ferromagnetic
$-1 < \chi < 0$	$0 < \chi < \epsilon$	$\chi \gg 1$
$0 \leq \mu_r < 1$	$1 < \mu_r < 1 + \epsilon$	$\mu_r \gg 1$
$\mu < \mu_0$	$\mu > \mu_0$	$\mu \gg \mu_0$

5.5.1 Diamagnetism

Diamagnetic substances are those which have tendency to move from stronger to the weaker part of the external magnetic field. In other words, unlike the way a magnet attracts metals like iron, it would repel a diamagnetic substance.

Figure 5.7(a) shows a bar of diamagnetic material placed in an external magnetic field. The field lines are repelled or expelled and the field inside the material is reduced. In most cases, this reduction is slight, being one part in 105. When placed in a non-uniform magnetic field, the bar will tend to move from high to low field.

The simplest explanation for diamagnetism is as follows. Electrons in an atom orbiting around nucleus possess orbital angular momentum. These orbiting electrons are equivalent to current-carrying loop and thus possess orbital magnetic moment. Diamagnetic substances are the ones in which resultant magnetic moment in an atom is zero. When magnetic field is applied, those electrons having orbital magnetic moment in the same direction slow down and those in the opposite direction speed up. This happens due to induced current in accordance with Lenz's law which you will study in Chapter 6. Thus, the substance develops a net magnetic moment in direction opposite to that of the applied field and hence repulsion. Some diamagnetic materials are bismuth, copper, lead, silicon, nitrogen (at STP), water and sodium chloride. Diamagnetism is present in all the substances. However, the effect is so weak in most cases that it gets shifted by other effects like paramagnetism, ferromagnetism, etc. The most exotic diamagnetic materials are superconductors. These are metals, cooled to very low temperatures which exhibits both perfect conductivity and perfect diamagnetism. Here the field lines are completely expelled! $\chi = -1$ and $\mu_r = 0$. A superconductor repels a magnet and (by Newton's third law) is repelled by the magnet. The phenomenon of perfect diamagnetism in superconductors is called the Meissner

effect, after the name of its discoverer. Superconducting magnets can be gainfully exploited in variety of situations, for example, for running magnetically levitated superfast trains.

5.5.2 Paramagnetism

Paramagnetic substances are those which get weakly magnetised when placed in an external magnetic field. They have tendency to move from a region of weak magnetic field to strong magnetic field, i.e., they get weakly attracted to a magnet. The individual atoms (or ions or molecules) of a paramagnetic material possess a permanent magnetic dipole moment of their own. On account of the ceaseless random thermal motion of the atoms, no net magnetisation is seen. In the presence of an external field B_0 , which is strong enough, and at low temperatures, the individual atomic dipole moment can be made to align and point in the same direction as B_0 . Figure 5.7(b) shows a bar of paramagnetic material placed in an external field. The field lines get concentrated inside the material, and the field inside is enhanced. In most cases, this enhancement is slight, being one part in 10^5 . When placed in a non-uniform magnetic field, the bar will tend to move from weak field to strong. Some paramagnetic materials are aluminium, sodium, calcium, oxygen (at STP) and copper chloride. For a paramagnetic material both χ and μ_r depend not only on the material, but also (in a simple fashion) on the sample temperature. As the field is increased or the temperature is lowered, the magnetisation increases until it reaches the saturation value at which point all the dipoles are perfectly aligned with the field.

5.5.3 Ferromagnetism

Ferromagnetic substances are those which get strongly magnetised when placed in an external magnetic field. They have strong tendency to move from a region of weak magnetic field to strong magnetic field, i.e., they get strongly attracted to a magnet. The individual atoms (or ions or molecules) in a ferromagnetic material possess a dipole moment as in a paramagnetic material. However, they interact with one another in such a way that they spontaneously align themselves in a common direction over a macroscopic volume called domain. The explanation of this cooperative effect requires quantum mechanics and is beyond the scope of this textbook. Each domain has a net magnetisation. Typical domain size is 1mm and the domain contains about 10^{11} atoms. In the first instant, the magnetisation varies randomly from domain to domain and there is no bulk magnetisation. This is shown in Fig. 5.8(a). When we apply an external magnetic field B_0 , the domains orient themselves in the direction of B_0 and simultaneously the domain oriented in the direction of B_0 grow in size. This existence of domains and their motion in B_0 are not speculations. One may observe this under a microscope after sprinkling a liquid suspension of powdered ferromagnetic substance of samples. This motion of suspension can be observed. Fig. 5.8(b) shows the situation when the domains have aligned and amalgamated to form a single 'giant' domain. Thus, in a ferromagnetic material the field lines are highly concentrated. In non-uniform magnetic field, the sample tends to move towards the region of high field. We may wonder as to what happens when the external field is removed. In some ferromagnetic materials the magnetisation persists. Such materials are called hard magnetic materials or hard ferromagnets. Alnico, an alloy of iron, aluminium, nickel, cobalt and copper, is one such material. The naturally occurring lodestone is another. Such materials form permanent magnets to be used among other things as a compass needle. On the other hand, there is a class of ferromagnetic materials in

which the magnetisation disappears on removal of the external field. Soft iron is one such material. Appropriately enough, such materials are called soft ferromagnetic materials. There are a number of elements, which are ferromagnetic: iron, cobalt, nickel, gadolinium, etc. The relative magnetic permeability is >1000 !

The ferromagnetic property depends on temperature. At high enough temperature, a ferromagnet becomes a paramagnet. The domain structure disintegrates with temperature. This disappearance of magnetisation with temperature is gradual.